

Multi-Agent Court-Act Assistant

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Problem statement



Open corpora of Russian courts of general jurisdiction (SOY) are large, mixed, and hard to search for analogous cases.

A case card stores metadata and sometimes HTML; a parallel corpus splits the act into header, facts (fabula), and operative part. There is no gold outcome label.

Legal LLMs often invent citations. Predicting a judgment from this data would be scientifically weak and practically misleading.

Scientific track: normalize the card into a Case Graph, retrieve comparable practice from an index, and draft a position that may cite only indexed case IDs.

Goal and tasks



Goal: build a working multi-agent prototype that analyzes a SOY case card, retrieves analogous practice, and drafts a grounded legal position.

Tasks:

1. Analyze the JSONL cards and parquet act-splits; define the Case Graph schema.
2. Implement ETL into SQLite so agents never scan raw gigabytes.

Implement a LangGraph pipeline with playbooks (dossier / analogs / risk / appeal / draft / full).

Evaluate routing, analog precision@5, and a citation Guard; ship CLI and Streamlit UI.

Related work



Zhong et al.; Chalkidis et al., LexGLUE / LEGAL-BERT: classification and judgment prediction. Gap: our corpus has no reliable outcome label; a “win probability” would overclaim.

COLIEE; Shao et al., BERT-PLI: learned case retrieval. Gap: we need an explainable ranker on Russian SOY acts, not an English encoder.

Lewis et al., RAG; Cui et al., ChatLaw: generate from retrieved text. Gap: generation must not invent case IDs.

Wu et al., AutoGen; LangGraph: LLM multi-agent stacks. We use an explicit linear graph: nodes outside the plan skip.

Magesh et al.: leading legal AI tools hallucinate citations. This motivates Guard against the SQLite index.

Takeaway: the gap is grounded retrieval and drafting – not another outcome classifier.

Methods and tools



Methods: rule-based legal NLP, Case Graph normalization, lexical information retrieval, multi-agent orchestration with playbooks.

Analog score (explainable): 5·same article + 3·same judge + 2·same region + 1.5·same instance + Jaccard overlap of fabula tokens.

Tools: Python 3.11+, LangGraph, Pydantic, PyArrow, SQLite (WAL, batch upsert), Streamlit; optional Ollama (OpenAI-compatible API).

Data: docs.json SOY cards; correct_df_splitting_text.parquet (200 parts, text_1 / text_2 / text_3).

Scientific novelty



The task is restated as Case Graph construction and analog retrieval, not judgment prediction.

A linear LangGraph with six playbooks: unused nodes skip, so one graph serves dossier, practice search, risk, appeal, and drafting.

Hybrid analog score is transparent (article / judge / region / instance / Jaccard) and runs without embeddings.

Guard accepts a draft only if cited case IDs exist in the index – a direct response to citation hallucination.

The pipeline is usable without an LLM key (rules + index statistics); the model is optional wording, not the source of facts.

Solution: data and index



JSONL cards and parquet rows are normalized into one Case Graph (id, court, judge, region, charges, timeline, fabula, operative part).

ingest writes SQLite: sources parquet / docs / both, limit, reset; existing parquet IDs are skipped; batch insert + WAL.

Full load: 1,862,811 cases from 200 parquet parts. Agents query case_store.sqlite only.

Corpus mix: civil 1,609,786; administrative 161,766; unknown 91,121; criminal 138 – so the system is not a criminal-outcome predictor.

Solution: agents and playbooks



Linear graph: Orchestrator → Parser → Timeline → Act Reader → Qualification → Retriever → Risk → Appeal → Draft → Guard → Respond.

- Playbooks: dossier (card only); analogs; risk; appeal; draft; full. Orchestrator picks by query rules; an explicit --playbook is never overridden.
- Retriever ranks by the hybrid score; Respond prints markdown only from AgentState; Guard checks cited IDs.
- Interfaces share the same functions: CLI (ingest / status / ask) and Streamlit UI.

Demo: Streamlit query

**Юрист:
мультиагентная
система**

Разбор карточек СОЮ, поиск аналогов,
оценка рисков, черновики позиций.

Состояние индекса

Дел в индексе: 1.862.811

`docs.json :`
`/Users/anasenichenkova/Documents/вуз/итмо/наука_юрист/docs.json`

`parquet :`
`/Users/anasenichenkova/Documents/вуз/итмо/наука_юрист/data/correct_df_splitted_text.parquet`

`store :`
`/Users/anasenichenkova/Documents/вуз/итмо/`

 **Задайте вопрос агенту**

Вопрос / запрос

найди похожие дела о взыскании налоговой недоимки

Плейбук

case_id

№ дела

Аналоги — подбор практики

0004a19c57c5e1b8ab09047870276c76

02-0001/2024

▶ Запустить агента

Index of 1,862,811 cases. Query: tax-arrears analog search. Playbook: analogs.

Demo: analog table and graph steps



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Состояние индекса

Дел в индексе: 1.862.811

docs.json :

/Users/anasenikova/Document
s/вуз/итмо/
научка_юрист/docs.json

parquet :

/Users/anasenikova/Document
s/вуз/итмо/
научка_юрист/data/correct_df_sp
litted_text.parquet

store :

/Users/anasenikova/Document
s/вуз/итмо/
научка_юрист/legal_multiagent/d
ata/case_store.sqlite

Загрузка данных

Источник

parquet + docs.json

Лимит карточек

Карточка: — Акт: —

Аналоги (20)

№ дела	id	Статья	Исход	Регион	Судья	Score	Прич
0bf9ad2ea5d5f346f5a1df856199eced	0bf9ad2ea5d5f346f5a1df856199eced	—	Производство прекращено	—	—	1	похож
2-416/11	0004a19c57c5e1b8ab09047870276c76	75	В иске отказано	—	ковалевой е.	0.923	похож
2-614/2010	012fcfb7984015602fe9b63afebd0648	397	В иске отказано	—	ежелого в.	0.91	похож
0b3224402736d919b276269ef522f0e7	0b3224402736d919b276269ef522f0e7	222	—	—	гвоздилиной и.	0.818	похож
13954b70ae65852ceff1f0b9b5c88cbc	13954b70ae65852ceff1f0b9b5c88cbc	—	Производство прекращено	—	чистякова о.	0.818	похож
0c35c74052708643d9a61e4167596bec	0c35c74052708643d9a61e4167596bec	357	Иск удовлетворён	—	—	0.757	похож
0af0179bc49821bc8c31a82f95c557fd	0af0179bc49821bc8c31a82f95c557fd	167	—	—	коржевой м.	0.711	похож
2-6373/10	04aae1378f888de58e1eb6ce8938a196	75	В иске отказано	—	зуевой в.	0.692	похож
07b023812b1d771dbf165b4f5fbb163e	07b023812b1d771dbf165b4f5fbb163e	75	Взыскание	—	архиповой н.	0.635	похож
2-2677/2017	0e825cbb60bef82df39eeb8ae63d0d50	194	—	—	—	0.635	похож

Шаги выполнения

orchestrator:analogs → parser:skip → timeline:skip → act_reader:skip → qualification → retriever → risk:skip → appeal:skip → draft:skip → guard:skip → respond

Сырые данные (JSON)

История запросов

1. найди похожие дела о взыскании налоговой недоимки — плейбук: `analogs`

20 ranked analogs. Unused LangGraph nodes skip (parser → ... → Guard).

10

Main results



Working artifact: legal_multiagent (CLI, Streamlit, README). Full index: 1.86M cases (~69 GB SQLite).

30 automated tests passed (normalization, retrieval, routing, Guard, Risk). No invented facts in Respond.

Retriever: $\geq 60\%$ of top-5 analogs share article and region. Guard rejects unknown case IDs.

Code and README: repository root (cognitive-funnel README; not a public license).

Kind	Records	Share	Role in system
Civil	1,609,786	86.4%	main analog pool
Administrative / other / criminal	253,025	13.6%	sparse; 138 criminal

Main results



Qualitative checks

Routing: five query types map to the intended playbook.

Respond: final answer contains only numbers and IDs present in state.

Risk: wording states this is not a court forecast; warns when analogs are few.

Limits of the current evaluation

No lawyer-labeled “this analog is useful” set on the full corpus (precision@5 is a proxy).

LLM branches (act reader / drafting) were not run e2e without an API key.

Lexical Jaccard can promote a different article if token overlap is high.

Expected next results



Fabula embeddings for analog search, compared with the current hybrid score.

A small lawyer-labeled relevance slice and precision/recall on that slice.

Mocks for Ollama so the full LangGraph path runs in CI.

Submit the Elsevier-format draft (paper/main.tex) to a student / legal-IR workshop.

Conclusion



The goal is met: a prototype that builds a Case Graph, retrieves analogs from SQLite, and drafts a position without citing missing IDs.

Task 1–2: schema + ETL for 1.86M parquet acts. Task 3: LangGraph playbooks, CLI, UI. Task 4: 30 tests; routing and Guard hold.

The system is an assistant for practice search, not a judgment classifier and not legal advice.

Next: embeddings, human relevance labels, and a workshop submission.

Publication readiness



A full manuscript is prepared in Elsevier elsarticle (paper/main.tex + references.bib).

Status: draft / preprint in the repository – not submitted, no acceptance yet.

Venue: to be selected (legal IR, multi-agent systems, or an ITMO student track) within three months.

No reviewer reports yet. The Overleaf-ready source can be compiled as pdfLaTeX.

Thank you
for your attention!

iT's **MO** *re than a*
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